Knowledge on **rare diseases** and **orphan drugs**

COVID-19 & Rare diseases Rare Diseases Resources for Refugees/Displaced Persons

Porphyria cutanea tarda

Suggest an update

Disease definition

A rare hepatic porphyria with cutaneous expression (PCT) characterized by bullous photodermatosis.

ORPHA:101330

Classification level: Disorder

Synonym(s):

PCT

Source: PubMed ID 30683557
31085196

Prevalence: 1-9 / 100 000

Inheritance: Autosomal dominant, Multigenic/multifactorial

Age of onset: Adult

ICD-10: E80.1

ICD-11: 5C58.10

OMIM: 176090
176100

UMLS: C0162566

MeSH: D017119

GARD: 7433

MedDRA: 10036183

Summary

Epidemiology

This is the most common form of cutaneous porphyria. Its prevalence in Western Europe is around 1/25,000. Men are more affected than women.

Clinical description

Occurring in adults, PCT is acquired (75% of cases) or familial (25% of cases). Manifestations generally appear earlier in familial cases. Some risk factors can trigger symptoms: excessive alcohol consumption, hepatitis C, estrogen intake, iron metabolism gene mutations with overload (hemochromatosis). The main clinical symptoms include extreme skin fragility, followed by bullous skin lesions on sun-exposed surfaces (hands, face). Healing is slow and often followed by hyper- and hypopigmentation. Skin lesions of varying ages are highly characteristic of the disease. These symptoms may be accompanied by hypertrichosis (especially facial) and, more rarely, by sclerodermiform lesions. Hepatopathy should be investigated (siderosis, steatosis, chronic inflammatory disorders, etc.).

Etiology

PCT is caused by a deficiency in uroporphyrinogen decarboxylase (UROD, the fifth enzyme in the heme biosynthesis chain). In the familial form of the disease, this deficiency is due to heterozygous mutations in the *UROD* gene (NM_000374.5), which codes for UROD and leads to an accumulation of porphyrins (uro- and hepta-carboxylic porphyrins) in the liver.

Diagnostic methods

Diagnosis is based on high concentrations of porphyrins in plasma, with a fluorimetric peak at 620 nm, and high concentrations of isocoproporphyrins in stools, which is specific to PCT. UROD deficiency in red blood cells confirms the diagnosis of familial PCT. In contrast, normal UROD activity in red blood cells is in favor of sporadic PCT. Skin biopsy is not very informative and is not recommended.

Differential diagnosis

Differential diagnosis mainly includes variegate porphyria (diagnosed on the basis of a characteristic fluorometric peak in plasma), hereditary coproporphyria, the photodermatosis of acute intermittent porphyria (in renal failure condition), and pseudo-porphyrias.

Antenatal diagnosis

Antenatal diagnosis is theoretically possible in familial forms, but is not offered.

Genetic counseling

The pattern of transmission for familial PCT is autosomal dominant, with low penetrance. Therefore, genetic counseling should be offered to those families to identify individuals likely to develop or transmit the disease.

Management and treatment

Management includes removal of triggers, management of underlying liver disease, phlebotomy and/or low-dose chloroquine (100 mg twice weekly) to achieve complete remission whether or not there is iron overload. A relapse is possible, and the treatment will be the same. Discontinuation of alcohol and estrogen-progestin therapy, and treatment of concomitant hepatitis C, are essential for remission.

Prognosis

PCT is not life-threatening and its prognosis is good. It is, however, a risk factor for the development of hepatocellular carcinoma.

Last update: ***March 2024*** *- Expert reviewer(s):* ***Dr Neila TALBI / MetabERN****

* European Reference Network

A definition / summary on this disease is available in Français, Español, Deutsch, Português, Nederlands

Detailed information

: produced/endorsed by ERN(s) : produced/endorsed by FSMR(s)

Guidelines

Emergency guidelines

Français (2016.pdf) - *Orphanet Urgences*

Polski (2007.pdf) - *Orphanet Urgences*

English (2007.pdf) - *Orphanet Urgences*

Español (2007.pdf) - *Orphanet Urgences*

Italiano (2007.pdf) - *Orphanet Urgences*

Português (2007.pdf) - *Orphanet Urgences*

Anesthesia guidelines

English (2014) - *Orphananesthesia*

Čeština (2014) - *Orphananesthesia*

Additional information

Further information on this disease

- Classification(s) (5)
- Gene(s) (2)
- Clinical Signs and Symptoms

Patient-centred resources for this disease

- Expert centre(s) (316)
- Networks of expert centre (15)
- Diagnostic tests (78)
- Patient organisation(s) (144)
- Federation/alliance(s) (48)
- Orphan designation(s) and orphan drug(s) (4)

Research activities on this disease

- Research project(s) (42)
- Biobank(s) (9)
- Registry(ies) (29)
- Network of experts (11)

Newborn screening

- Newborn screening library

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